Presentation of the Research

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- Writing up the research
- Written Presentation
  - Academic papers
  - Theses and dissertations
  - Technical reports
- Visual Presentations
  - Posters
  - Infographics
  - Demonstration
- Oral Presentation
  - Conference talks
  - Thesis defense
- Conferences and Journal

Presentation of the Research

- The final "P" of research is *Presentation*, which focuses on effectively communicating your work to the academic community for scrutiny and evaluation.
  - purpose, products, process, participants and paradigm
- **Importance of Presentation Skills**: Beyond conducting research, researchers must develop skills in writing and speaking about their work in an accepted academic style.
- **Forms of Academic Presentation**: Research can be presented through a **thesis, journal article, or conference paper**, each tailored to **specific academic audiences**.
- **Alternative Presentation Methods**: Work can also be showcased via conference posters, software demonstrations, or exhibitions, offering diverse ways to engage with the community.
- **Oral Defense (Viva)**: Students are often required to discuss and justify their research during a viva, an oral examination with examiners.

Writing Up the Research

- **Written Account as Tangible Evidence**:
  - For most researchers, a *written account* is the primary means of evaluation by others.
  - It serves as the **key documentation** of the research process, findings, and conclusions.
- **Contextualizing Creative Work**:
  - For design and creation researchers (e.g., digital art or computer-based systems), a *textual report* is essential.
  - It **describes, justifies, and contextualizes** the work, ensuring it is understood and evaluated within the academic framework.

Getting Started with Writing

- **Writing Clarifies Ideas**:
  - Writing helps researchers clarify their thoughts and discover their true perspectives.
  - Keeping private notes during the research process is a good practice to track ideas, problems, and interpretations.
- **Avoid Last-Minute Writing**:
  - Leaving all writing until the end can lead to time constraints and a rushed, lower-quality thesis.
  - Unlike undergraduate assignments, research reports require multiple drafts over weeks or months.
- **Prevent Overwhelm**:
  - Writing during the research process reduces the burden of organizing large amounts of data and papers at the end.
  - Many students fail to complete their work because they struggle to compile everything into a coherent thesis.
- **Writing as a Core Research Skill**:
  - Writing is an integral part of research, requiring practice, feedback, and revision.
  - Regularly sharing and rewriting your work improves your writing proficiency.
- **Early Writing Opportunities**:
  - The literature review can be written early while the material is fresh in your mind.
  - Stay updated with new publications to ensure your review remains current.
- **Tackle Methodology Early**:
  - The methodology chapter/section is often easier to write early, as it reflects what you've done and why.
  - This section provides a clear foundation for the rest of your thesis or report.

Structure and Style of Academic Writing

- **Conventional Structure**:
  - **Beginning**: Title, authors, abstract, keywords (for search engine indexing).
  - **Main**: Introduction, literature review, methodology, results, discussion, limitations, conclusions, recommendations, and suggestions for further work.
  - **End**: References and appendices (e.g., questionnaires, interview transcripts, program code).
- **Acknowledgements**:
  - Acknowledge contributors (supervisors, participants, colleagues, funding bodies).
  - Placement: In a thesis, after the abstract; in journal articles, before references.
- **Quantitative vs. Qualitative Research**:
  - **Quantitative**: Separate sections for results and discussion.
  - **Qualitative**: Integrate data (e.g., interview quotes) and interpretation in one section.
- **Design and Creation Research**:
  - Unique structure: Introduction, literature review, development stages (analysis, design, implementation, testing), evaluation, and conclusions.
  - Include key code segments in the main text; place full code in appendices with clear documentation.
- **Signposting and Headings**:
  - Use meaningful headings and subheadings to guide readers.
  - Provide signposts (e.g., "The last chapter explained... This chapter covers...").
- **Writing Style**:
  - Traditionally, third-person passive was used (e.g., "An experiment was designed...").
  - Increasingly, first-person active is acceptable (e.g., "We designed an experiment...").
  - Avoid "I" for single researchers; use "the author" or consult discipline-specific norms.

Making an Argument in Research Writing

- **Research as an Argument**:
  - A thesis or research paper is essentially an argument supported by evidence.
  - The goal is to convince readers that you've created new knowledge through an appropriate academic process.
- **Key Questions to Guide Your Argument**:
  - What is the research question?
  - What is the conceptual framework?
  - How was the question answered?
  - What is the answer?
  - What evidence supports the answer?
  - **So what?** (Why does the answer matter? Does it fill a gap, extend a model, or have practical implications?)
- **Logical Structure and Style**:
  - Ensure the argument has a clear, logical sequence.
  - Follow conventional academic style and structure to meet reader expectations.
  - Avoid spelling and grammar errors to maintain credibility.
- **Supporting Evidence**:
  - Base assertions and conclusions on evidence, not assumptions.
  - Use literature reviews, empirical data, and analysis to substantiate claims.
- **Presenting Data Effectively**:
  - For **quantitative data**: Use tables, graphs, and statistical analysis.
  - For **qualitative data**: Include direct quotes, matrices, and visual aids.
  - Ensure all tables and figures are clearly labeled and easy to understand.
- **Referencing and Precision**:
  - Use the referencing style required by your institution, journal, or conference.
  - Avoid common word confusions (e.g., affect/effect, practice/practise) to maintain clarity and professionalism.

Developing a Writing Routine

- **Find Your Writing Style**:
  - Experiment with different approaches: daily word targets, longhand vs. typing, or writing in bursts of inspiration.
  - Choose an environment that suits you (e.g., background music or silence).
- **Optimize Your Productivity**:
  - Write new content during your most productive time of day.
  - Use less productive times for mechanical tasks like spell-checking and reference verification.
- **Focus on Progress, Not Perfection**:
  - The first draft doesn't need to be perfect; aim to produce words and refine later.
  - Use a word processor to easily edit sentences, move sections, and improve drafts.
- **Overcome Writer's Block**:
  - Write imperfectly: jot down close words or fragments if full sentences don't come.
  - If stuck, switch to a different section or take a walk to clear your mind.
- **Flexibility in Writing Plans**:
  - Don't rigidly follow your plan; write what feels most motivating at the time.
  - Adapt your routine to maintain momentum and creativity.
- **End on a High Note**:
  - Finish your writing session mid-sentence or mid-paragraph to make it easier to resume later.
  - This trick helps maintain continuity and motivation for the next session.

Reviews and Feedback

- **Self-Review After Completion**:
  - Leave your thesis chapter or paper for a few days before revisiting it.
  - This helps identify unclear structure, missing evidence, or wordy prose.
- **Peer Feedback**:
  - Have a friend or colleague read your text to spot mistakes and unclear sections.
  - Fresh eyes can catch issues you might overlook due to familiarity with the material.
- **Supervisor's Role**:
  - Share a polished draft with your supervisor, not the first draft.
  - Utilize their experience to improve the structure, clarity, and academic rigor of your work.
- **Conference/Journal Reviews**:
  - Expect 2-3 anonymous reviews when submitting to conferences or journals.
  - Rarely is a paper accepted without revisions; reviews often highlight strengths, weaknesses, and suggest improvements.
- **Handling Reviewer Feedback**:
  - Be prepared for minor edits or substantial rewrites, including additional empirical research.
  - Even if rejected, constructive feedback can guide future improvements.
- **Emotional Response and Growth**:
  - Initial disappointment with critical reviews is normal, even for experienced researchers.
  - Reflect on feedback and recognize its value in enhancing the quality and impact of your work.

Thesis vs Dissertation



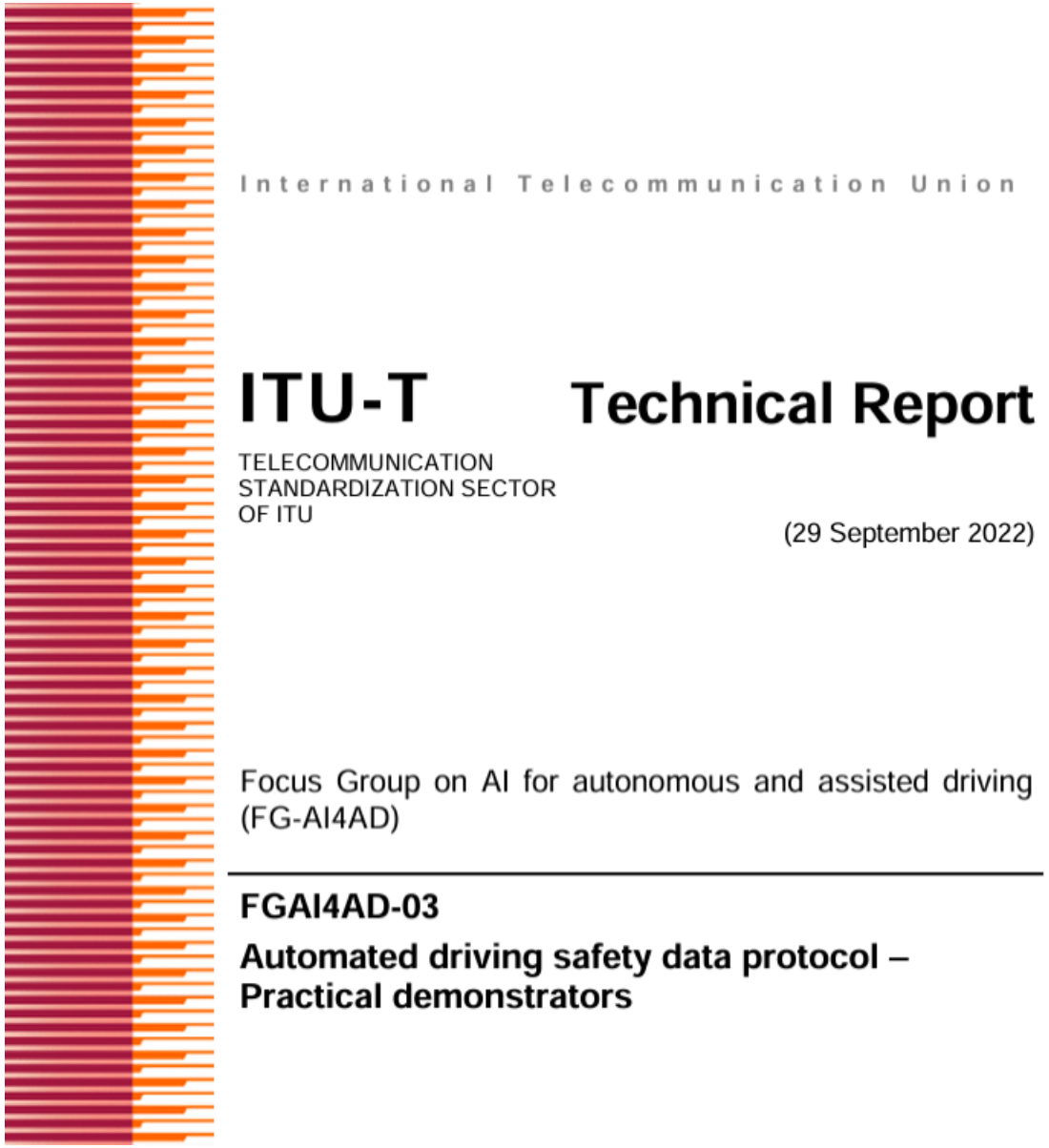
- **Purpose and Scope**:
  - **Thesis**: Demonstrates mastery of a subject, often required for a master's degree.
  - **Dissertation**: Contributes new knowledge to the field, required for a doctoral degree.
- **Research Focus**:
  - **Thesis**: Involves analyzing existing research and presenting original insights.
  - **Dissertation**: Requires conducting original research and performing in-depth analysis.
- **Length and Depth**:
  - **Thesis**: Typically shorter (50-100 pages), focusing on a specific topic.
  - **Dissertation**: Longer (100-300+ pages), exploring a broader research question.
- **Originality**:
  - **Thesis**: Builds on existing research with some original analysis.
  - **Dissertation**: Demands significant originality, often leading to new theories or findings.
- **Defense Process**:
  - **Thesis**: May or may not require a formal defense, depending on the program.
  - **Dissertation**: Almost always requires a formal defense before a committee.
- **Outcome**:
  - **Thesis**: Demonstrates the ability to synthesize and analyze information.
  - **Dissertation**: Establishes the candidate as an expert in their field, contributing to academic knowledge.

Technical Report

- "A technical report is a document that describes the process, progress, or results of technical or scientific research or the state of a technical or scientific research problem. It might also include recommendations and conclusions of the research."

https://en.wikipedia.org/wiki/Technical_report

- A technical report is a detailed document that communicates the results, processes, and findings of technical or scientific research.
- Used to provide a clear, structured, and comprehensive account of research or project work. Often used in engineering, science, and industry to share information with stakeholders.



Technical Report

Purpose:

- **Title Page**: Includes the title, author, date, and institution/organization.
- **Abstract**: A brief summary of the report's purpose, methods, results, and conclusions.
- **Introduction**: Explains the problem, objectives, and scope of the report.
- **Methodology**: Describes the methods, tools, and procedures used in the research.
- **Results**: Presents the findings, often with tables, graphs, and charts.
- **Discussion**: Analyzes the results, interprets their significance, and addresses limitations.
- **Conclusion**: Summarizes key findings and their implications.
- **Recommendations**: Suggests actionable steps or future research directions.
- **References**: Lists all sources cited in the report.
- **Appendices**: Includes supplementary material (e.g., raw data, calculations, code).

- **Audience**:
  - Targeted at technical or specialized audiences, such as researchers, engineers, or industry professionals.

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- **Key Characteristics**:
  - **Clarity**: Written in precise, straightforward language.
  - **Accuracy**: Data and findings must be accurate and verifiable.
  - **Objectivity**: Presents information without bias or subjective opinions.
  - **Visual Aids**: Uses tables, graphs, and diagrams to enhance understanding.
- **Use Cases**:
  - Documenting experiments, research projects, or technical investigations.
  - Sharing findings with stakeholders, funding agencies, or regulatory bodies.
  - Serving as a reference for future research or decision-making.
- **Design Tips for Technical Reports**:
  - Use headings and subheadings for easy navigation.
  - Incorporate visuals (e.g., charts, graphs, diagrams) to illustrate data.
  - Maintain a professional layout with consistent fonts, spacing, and formatting.
  - Include a table of contents for longer reports.

Conference Paper Presentations

- **Attendance and Presentation Requirements**:
  - If your paper is accepted, you must attend the conference and present it (only one author for multi-author papers).
  - Papers are included in proceedings only if presented, so ensure you have funding for travel, accommodation, and fees before submitting.
- **Applicability to In-House Presentations**:
  - The guidelines for conference presentations also apply to presenting your work at your own institution.
  - Focus on clear communication, engaging delivery, and effective use of visuals for both settings

Preparing the Presentation

- **Time Allocation**:
  - Typically, presentations are 20 minutes long with 10 minutes for Q&A, but confirm the exact time with the conference organizers.
- **Avoid Reading the Paper**:
  - Do not read your paper verbatim; the audience can read it themselves in the proceedings.
  - Use slides (e.g., PowerPoint) to highlight key points and talk around them.
- **Slide Content**:
  - Include relevant visuals like screenshots, webpages, or software interfaces to enhance understanding.
  - Limit slides to 10-14 for a 20-minute presentation to avoid rushing or overwhelming the audience.
- **Focus on Key Aspects**:
  - Highlight 1-2 main points, such as:
    - What you're proud of.
    - A key argument.
    - Audience interests.
    - Areas where you seek feedback.
- **Font Size and Handouts**:
  - Use a font size of at least 24 points for readability.
  - For complex diagrams, provide handouts so the audience can follow along.
- **Alternative Presentation Methods**:
  - Overhead projectors and transparencies are less common but offer flexibility.
  - Confirm availability with organizers if using this method.

Getting Ready to Speak

- **File Preparation and Naming**:
  - Bring your presentation on a storage medium (e.g., USB, CD) and ensure compatibility with the conference system.
  - Name your file uniquely, starting with your name (e.g., Watson_WWW_Conference_05.ppt), to avoid confusion.
- **Loading Your Presentation**:
  - Load your file onto the conference system during breaks or at the start of the day.
  - Seek technical assistance if needed and verify your file is accessible.
- **Familiarize Yourself with Equipment**:
  - Test the computer system, including page-up/down keys or touchpad, before your session.
  - Practice with unfamiliar devices (e.g., touchpads) beforehand to avoid issues.
- **Using Your Own Laptop**:
  - Confirm with organizers if using your laptop is allowed.
  - Ensure proper connections (e.g., network, projector) and assist the next speaker after your presentation.
- **Arrival and Setup**:
  - Arrive at least 10 minutes early to your session.
  - Introduce yourself to the session chair and other speakers, and position yourself for easy access to the podium.
- **Technical and Professional Etiquette**:
  - Be prepared for technical setups and ensure smooth transitions between speakers.
  - Maintain professionalism by assisting others and staying organized.

Doing the Presentation

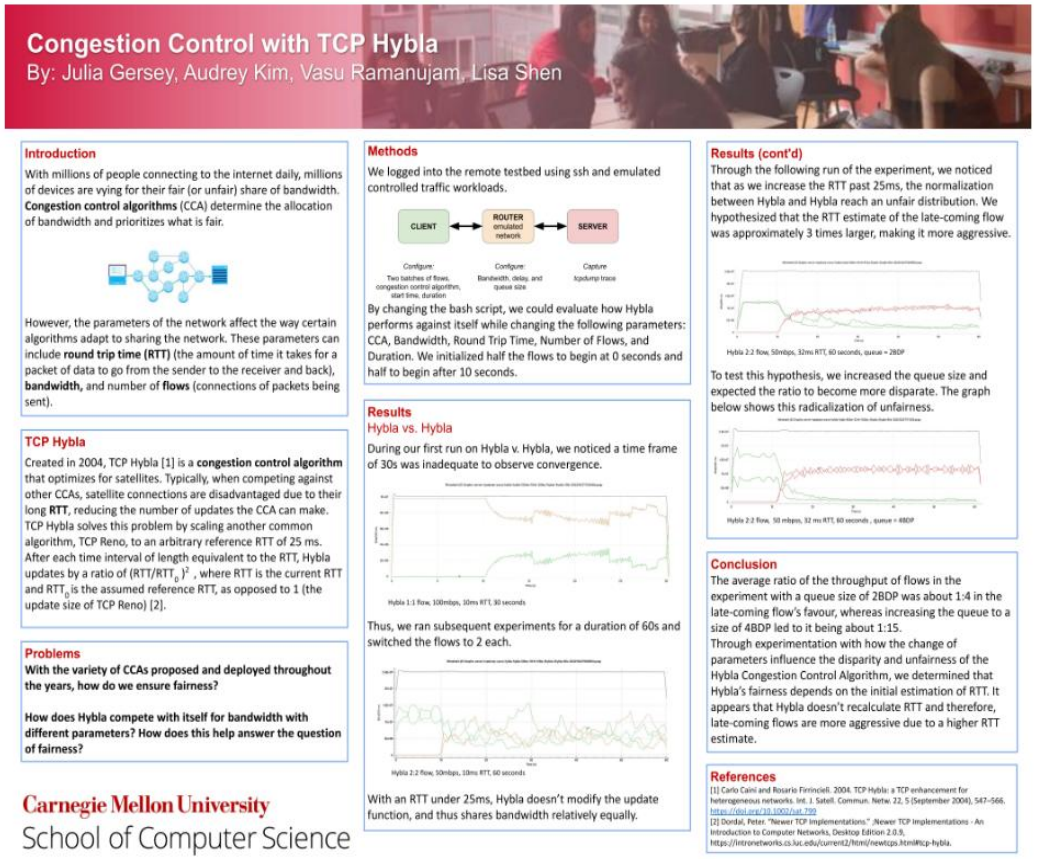
- **Delivery Tips**:
  - Speak slowly, clearly, and loudly to ensure the audience can follow.
  - Use a printed copy of your slides with notes as prompts to stay on track.
- **Engage with the Audience**:
  - Avoid constantly looking at the screen, notes, or computer.
  - Maintain eye contact with different audience members to build connection.
- **Handling Nervousness**:
  - Acknowledge your nervousness if it's your first presentation; audiences are usually sympathetic.
  - Remember, even experienced presenters feel nervous.
- **Time Management**:
  - Respect the allotted time; the session chair will signal when time is running out.
  - Avoid overrunning to ensure fairness to other presenters and the audience.
- **Audience Behavior**:
  - Don't be discouraged if people arrive late or leave early; it's common at conferences.
  - Stay for the entire session out of respect for fellow presenters.
- **Handling Questions and Discussants**:
  - If no questions are asked, the session chair may step in with prepared questions.
  - For difficult questions, admit if you don't know the answer and suggest discussing it later.
  - Use your final slide to guide audience questions and manage responses effectively

If you don't know how to answer a question, don't be afraid to say so. Try one of the

- 'That's an interesting aspect I hadn't thought of. But I'd like more time to think about it, so I'm not going to be able to respond right now. Sorry.'
- 'Thank you. That's a useful suggestion. Can you give me some references on that later?'
- I'd like to discuss that point with you in much more detail, but now probably isn't the time. Perhaps over coffee later?'

Posters and Exhibitions

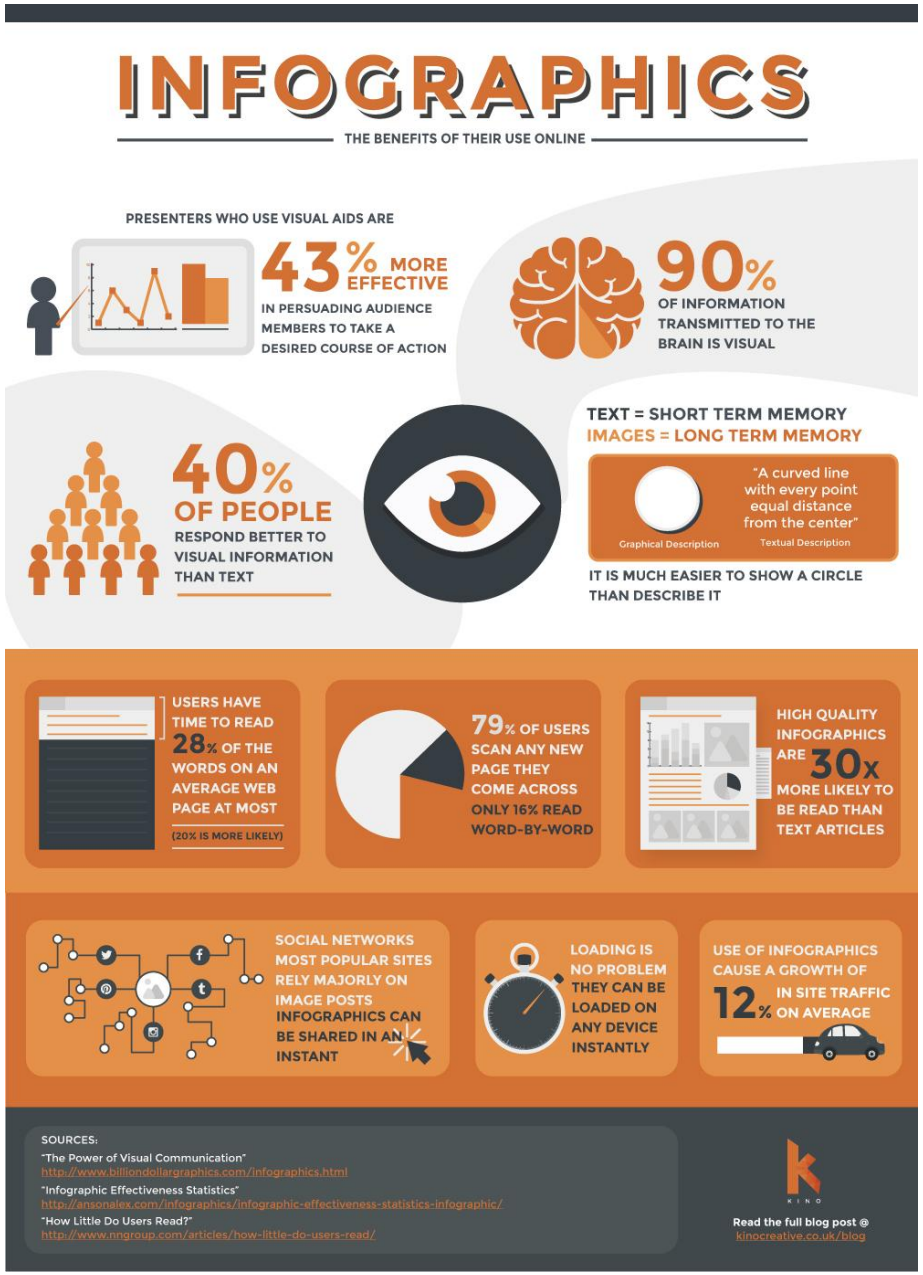
- **Poster Presentation Opportunity**:
  - Conferences may offer poster displays as an alternative to paper presentations.
  - Posters are displayed in a dedicated room, and you'll have scheduled times to discuss your work with delegates.
- **Poster Specifications**:
  - Check the allocated space (e.g., one A1 sheet or more) and design accordingly.
  - Posters can range from simple PowerPoint slides printed and mounted to professionally printed, high-quality designs.
- **Design Tips for Posters**:
  - Use eye-catching visuals and minimal text to summarize your research.
  - Ensure the title is clear, the content is legible from a distance, and include your full contact details.
- **Digital Art Exhibitions**:
  - Provide short textual explanations for each piece, readable from a distance.
  - Write a personal statement reflecting on your work's ideas and influences for the exhibition catalogue.
- **Engagement with Visitors**:
  - Attend pre-arranged times to discuss your work with exhibition visitors.
  - Be prepared to explain your research or artwork in detail and answer questions.
- **Professional Assistance**:
  - Seek help from your university's publicity department for creating professional-looking posters.
  - Ensure your presentation aligns with the conference or exhibition guidelines.



Team 2: Srini Seshan (mentor) Julia Gersey, Audrey Kim, Vasu Ramanujam & Lisa Shen

Infographics

- **Visual Communication**:
  - Infographics simplify complex research data into visually appealing, easy-to-understand formats.
  - Ideal for presenting trends, comparisons, and key findings at a glance.
- **Enhanced Engagement**:
  - Visual elements like charts, graphs, and icons capture attention better than text-heavy content.
  - Increase audience interest and retention of information.
- **Effective Storytelling**:
  - Infographics help narrate the research process, from problem identification to conclusions.
  - Use a logical flow to guide the audience through the data.
- **Versatility**:
  - Suitable for various platforms: conference presentations, journal articles, social media, and websites.
  - Adaptable to different research fields (e.g., science, social sciences, business).
- **Improved Accessibility**:
  - Break down technical jargon and make research accessible to non-expert audiences.
  - Useful for policymakers, industry professionals, and the general public.
- **Time-Saving**:
  - Summarize lengthy research papers or datasets into concise, digestible visuals.
  - Save time for both researchers and readers by highlighting key insights quickly.
- **Design Tips for Effective Infographics**:
  - Use a clear hierarchy (headlines, subheadings, visuals).
  - Choose a consistent color scheme and fonts.
  - Incorporate charts (bar, pie, line) and icons for better representation.
  - Keep text minimal and focus on visuals to convey the message.



Software Demonstrations

- **Pre-Demonstration Preparation**:
  - Ensure your software runs flawlessly on the designated machine.
  - Check for required plug-ins, database versions, or updates well in advance.
- **Laptop Usage**:
  - If using a laptop, ensure the battery is fully charged.
  - Verify institutional policies on connecting to power supplies for safety compliance.
- **Setup Before Arrival**:
  - Have all files open and programs running before the audience or examiners arrive.
  - Avoid delays by preparing the system in advance (e.g., booting up, logging in).
- **Structured Presentation**:
  - Demonstrate your software in a logical, step-by-step manner.
  - Avoid rushing or randomly showcasing features; focus on key functionalities (e.g., creating, modifying, deleting records).
- **Handling Malfunctions**:
  - If the software fails, remain calm and avoid technical troubleshooting during the demo.
  - Apologize, mention it worked previously, and move to another part of the program.
- **Backup Plan**:
  - Be prepared to explain what the software should have done if it malfunctions.
  - Keep the discussion clear and concise, focusing on the intended functionality.

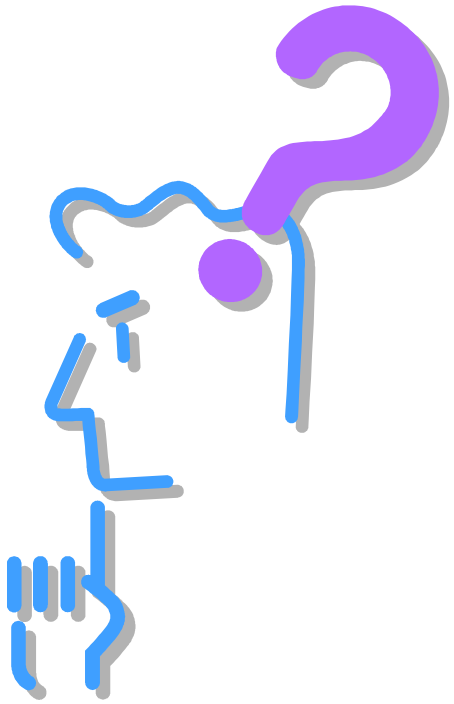
Preparing for Vivas

- **Viva Format Variations**:
  - In the UK: Private viva with internal and external examiners, lasting 1-3 hours.
  - In some countries: Public viva with a panel of examiners, possibly from different disciplines.
- **Purpose of the Viva**:
  - An opportunity to explain and justify your thesis work.
  - Demonstrate your understanding and contribution to the field.
- **Preparation Strategy**:
  - Re-read your thesis as if you were an examiner.
  - Anticipate potential questions and prepare clear, concise answers.
- **Common Examiner Questions**:
  - What debates/issues/problems does your research address?
  - Why did you choose this research strategy/methods/case?
  - What would you do differently if starting again?
- **Questions on Contribution**:
  - What is your main contribution to knowledge?
  - What did you enjoy most in your work?
  - How would you turn your thesis into journal articles?
- **Authorship Verification**:
  - Be prepared to explain the overall structure of your thesis.
  - Answer detailed questions about specific sections (e.g., "On page 72, you say…").
- **Broader Context Questions**:
  - How does your topic fit into current research trends?
  - Be ready to discuss the relevance and impact of your work.
- **Handling Criticism**:
  - Acknowledge limitations in your work and justify your choices.
  - Reference specific sections of your thesis to support your answers.
- **Responding to Unfamiliar Citations**:
  - If asked about an unknown author, remain calm and ask for clarification.
  - Example: "I'm not familiar with their work. How would it strengthen or challenge my thesis?"
- **Mock Viva Practice**:
  - Participate in or observe a mock viva to gain insight into the process.
  - Use the experience to refine your responses and build confidence.
- **Typographical Errors**:
  - Note any errors in your thesis before the viva.
  - Acknowledge them if pointed out by examiners to show awareness.
- **Enjoy the Experience**:
  - While nerve-wracking, most students find the viva enjoyable.
  - It's a chance to discuss your work as the leading expert in your topic.

Conference vs Journal Paper

- **Purpose and Scope**:
  - **Conference Papers**: Focus on presenting preliminary or ongoing research, often with a narrower scope.
  - **Journal Papers**: Provide comprehensive, finalized research with in-depth analysis and broader implications.
- **Length and Detail**:
  - **Conference Papers**: Shorter (typically 4-10 pages), concise, and focused on key findings.
  - **Journal Papers**: Longer (often 20+ pages), detailed, and include extensive literature reviews, methodologies, and discussions.
- **Review Process**:
  - **Conference Papers**: Undergo a quicker, less rigorous peer-review process.
  - **Journal Papers**: Subject to a more thorough and time-consuming peer-review process.
- **Timeliness of Publication**:
  - **Conference Papers**: Published quickly, often within months of submission.
  - **Journal Papers**: Take longer to publish due to the extensive review and revision process.
- **Audience and Dissemination**:
  - **Conference Papers**: Presented to a live audience at conferences, allowing for immediate feedback and networking.
  - **Journal Papers**: Reach a wider, global audience through academic journals, but lack direct interaction.
- **Content Focus**:
  - **Conference Papers**: Emphasize novelty, innovation, and recent developments.
  - **Journal Papers**: Focus on thoroughness, validation, and contribution to the field.
- **Format and Structure**:
  - **Conference Papers**: Follow a standardized format (e.g., abstract, introduction, methodology, results, conclusion).
  - **Journal Papers**: Include additional sections like detailed literature reviews, extensive data analysis, and future work.
- **Impact and Citations**:
  - **Conference Papers**: Often cited less frequently but provide early exposure to new ideas.
  - **Journal Papers**: More likely to be cited and considered authoritative sources.
- **Revision Opportunities**:
  - **Conference Papers**: Limited opportunities for revision before publication.
  - **Journal Papers**: Multiple rounds of revisions based on reviewer feedback.
- **Presentation vs. Publication**:
  - **Conference Papers**: Primarily presented orally or via posters at conferences.
  - **Journal Papers**: Published in print or online, accessible to a wider audience.
- **Target Audience**:
  - **Conference Papers**: Aimed at researchers and professionals attending the conference.
  - **Journal Papers**: Targeted at the broader academic community and practitioners in the field.
- **Career Impact**:
  - **Conference Papers**: Useful for early-career researchers to gain visibility and feedback.
  - **Journal Papers**: Essential for establishing credibility and contributing to academic advancement.

The questions to ask yourself first…



Is the paper worth writing?

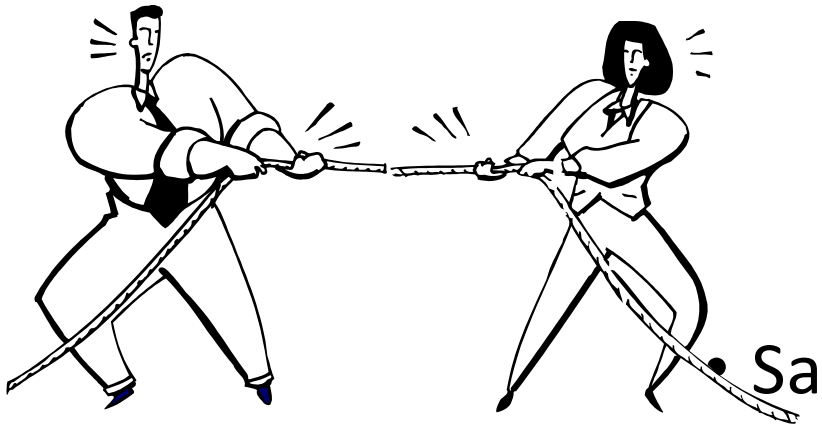
- What's in the literature?
- "So What?"
- It's a lot of work (average 20-30 drafts). Don't do it unless its worth it.

What do I have to say?

- A single question clearly stated with adequate evidence for the answer.
- Try stating the question and its answer in one simple sentence.

Is it one or more papers?

- Putting too much in one paper makes it diffuse and less compelling than if its focused



- Salami Science?

OK, So you want to/need to write a paper --> next questions

a. What is the right format for the message (original article/review?)

b. What is the right audience—who cares?

c. What journal should I choose?

Which journal?

- i. Is topic of my paper within its scope and format?
- ii. Would it match my audience?
- iii. Ask mentor or other senior researchers: appropriateness
- iv. Impact Factor
- v. Consequences of wrong decision: time lost; failure to publish

A scientific article as a critical argument

a. Statement of problem; posing a question
b. Presentation of evidence
c. Assessment of the validity of the evidence in the face of ..
    a. strengths/weaknesses
    b. other evidence
d. Conclusions

Literature Search First

- What has been done and what can you say that’s new?
- Be thorough in your search:---a high sensitivity/low specificity search.

The Title Page: Do it early—title; authors and their order; sources of support

a. Establishes responsibilities in paper writing
b. Avoids hurt feelings

The Process of Paper Writing

- Create an outline first
- Plan on multiple drafts:
  - Filename with dates
  - One filename written over with new draft
- Tables/figures early: prompt more analysis
- Deadlines for you and coauthors

The Introduction

- Draw audience in; be provocative
- Target journal specific audience
- Identify gaps in knowledge
- End with question/hypothesis



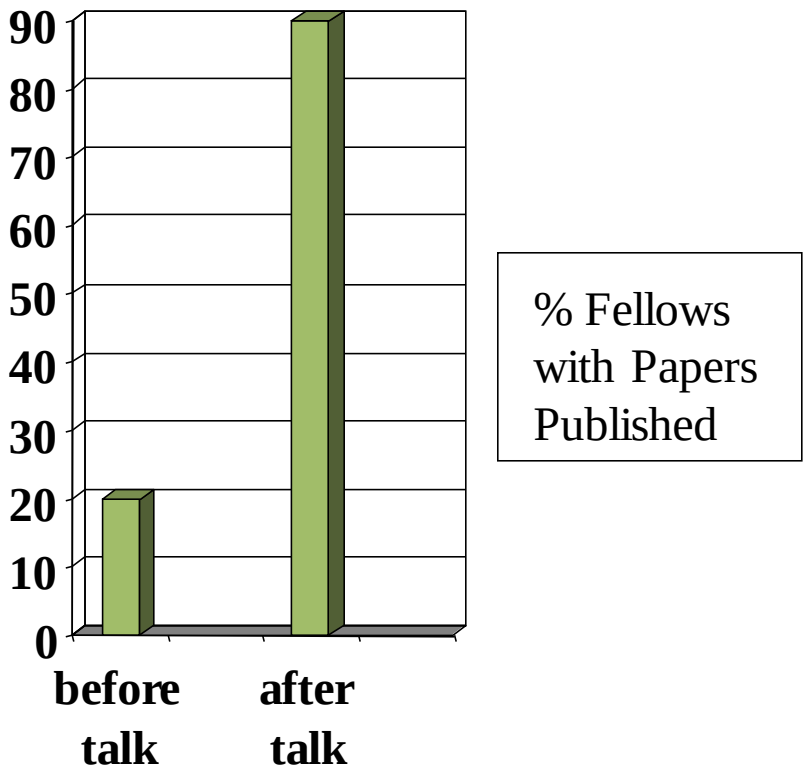
Knee Pain, Why?

The Methods Section

- Should include subheadings
- Write clearly enough to be understood by nontechnical reader and be replicated
- Past tense

The Results Section

- Organize around tables/figures
- Present tabular results selectively in text
- Past tense
- No interpretation; just the facts!
- Tables should stand on their own



The Discussion Section

- 1st paragraph: answer question/hypothesis
- Remainder:
  - Evidence pro and con: literature review
  - Strengths/limitations of your study
  - Implications of findings (be conservative)
  - Other findings of your study
- Last Paragraph: conclusion

Getting Tense!
PAST and PRESENT

- When quoting *previously published work*, refer to it in *present tense* (e.g. penicillin treats strep throat)

- When describing your *own study*, refer to work in *past tense* (e.g. we tested a new antibiotic for strep throat)

Pop Quiz: The good scientific paper …

A. Is focused on a specific question(s).
B. Covers a broad spectrum of disease or methodologic questions

- Less is More.

Pop Quiz: In a good scientific paper …

A: Abstract and tables and figures are understandable without reading whole paper.

B. Abstract and tables and figures are understandable only with reading whole paper.

Pop Quiz: In a good scientific paper ...

A: Writing is in passive voice (e.g. it was found that…).

B. Writing is in active voice (we found that…).

C. Writing mixes active and passive voice.

Pop Quiz: In a good scientific paper …

A: A term defined in the methods section is used again and again (a rose, a rose, a rose)

B. Various synonyms for a term are used to prevent reader boredom. (a rose, a flower with a thorny stem, a fragrant flower)

- Define a term and use it consistently. Otherwise, you'll confuse the reader.

Pop Quiz: In a good scientific paper ...

A: Writing is flowery

B. Writing is concise

- Generally, the shorter, the better

Proofread before Submitting

- ✓Are terms used consistently throughout?
- ✓Do numbers in abstract match numbers in text and tables?
- ✓Do citations in text match references?
- ✓Are Syntax and Grammar acceptable

A brief synopsis of writing an abstract

- It's a minipaper:
  - Introduction (usually 1-2 sentences)
  - Methods (often longest part)
  - Results
  - Discussion is limited to concluding statement
- Like a paper, requires many drafts, most oriented to presenting argument concisely

Getting the Reviews of Your Paper

- “The reviewer is always right.” (whether they are or not!)
- Don’t respond quickly. Digest reviews.

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If your paper was rejected…

- Was it sent out for review? If not, consider changing type of journal
- If reviews don't suggest changes, send it out quickly to another journal
- The 3 journal rule.

